FABRIZIO RUSSO

Research Associate, Imperial College London

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SUMMARY

Research Associate at Imperial College London, starting an AI in Policy Fellowship this coming July, with a PhD in Safe & Trusted AI, developing explainable, interactive causal discovery methods for high-stakes decision-making. As a key contributor to an ERC-POC grant built on my thesis, I'm now working towards applying my research in finance, healthcare and policymaking. I've published at UAI, AAAI, KR, ECAI, IJCAI, and CLeaR. With six years' experience consulting and leading data-science teams in banking, I bridge academia, industry and regulatory bodies to drive human-centred AI innovation.

WORK EXPERIENCE

Research Associate
Imperial College London
Full-time: Jan 2025-present
UK

- ERC Proof of Concept grant: Causal Argumentative Learning Assistant (CArLA)
- Research in Causal Discovery, Argumentation and Explainability

Graduate Teaching Assistant
Imperial College London
Part-time: Oct 2021 & 2022
UK

- Teaching support for *Introduction to Machine Learning* course

Freelance Consultant
Imperial College Consultants (ICON)
Part-time: Mar 2022 – Sep 2022
UK & USA

- Research on outlier identification and forecast for UK commercial properties index

Doctoral Researcher
Ernst & Young (in collaboration with STAI CDT)
Part-time: Jan 2021 – Apr 2021
UK

- Research on back-door poisoning attacks identification for Large Language Models

Head of Data Science
4most Europe Ltd
Full-time: Feb 2018 – Sep 2020
UK & Germany

- Replicate underwriter behaviour using machine learning
- Commercial credit risk scoring using machine learning
- Design, prototype and build an interactive monitoring tool (Tech of the Year - National Credit Awards '22)

Managing Consultant
4most Europe Ltd
Full-time: Jul 2016 – Jan 2018
UK & Canada

Credit Risk Consultant
4most Europe Ltd
Full-time: Jul 2015 – Jun 2016
UK

Credit Risk Analyst
General Electric Capital - UK Home Lending
Full-time: May 2014 – Jun 2015
UK

EDUCATION

PhD in Safe and Trusted Artificial Intelligence
Full-time: Oct 2020 – Feb 2025

Imperial College London
UK

- Supervisor: Prof. Francesca Toni
- Thesis: Causal Discovery for Trustworthy Artificial Intelligence
- Visiting PhD Student at King's College London (STAI CDT)

MSc in Applied Statistics (Distinction)
Part-time: Oct 2016 – Oct 2018

University of London - Birkbeck College
UK

- Dissertation: Macroeconomic Modelling through VAR and VEC

BSc in Economics of Financial Markets and Institutions (1:1)
Full-time: Oct 2009 – Oct 2012

University of Rome - Tor Vergata
Italy

- Dissertation: The Tobin Tax between Theoretical Wherewithal and Practical Feasibility

PUBLICATIONS

Conference Proceedings

- Li, Zihao and **Fabrizio Russo** (2026). "Leveraging Large Language Models for Causal Discovery: a Constraint-based, Argumentation-driven Approach". In: *42nd Conference on Uncertainty in Artificial Intelligence, UAI 2026 (to appear).*
- **Russo**, **Fabrizio** and Mark Somers (2026). "Operationalising Relative Causal Knowledge: Backbone Identifiability from Private Reports". In: *JoeFest: A Workshop in Honor of Joseph Y. Halpern, co-located with KR / FLoC 2026 (to appear).*
- **Russo**, **Fabrizio**, Giuseppe Mazzotta, Carmine Dodaro, Francesco Ricca, and Francesca Toni (2026). "Optimal Correction Sets for Argumentative Causal Discovery". In: *LINDA 2026: Logical Approaches to Handling Inconsistent Data, co-located with KR / FLoC 2026 (to appear).*
- Gehlot, Preesha, Anna Rapberger, **Fabrizio Russo**, and Francesca Toni (2026). "Heterogeneous Graph Neural Networks for Assumption-Based Argumentation". In: *Proceedings of the 40th Annual AAAI Conference on Artificial Intelligence, AAAI-26.*
- – (2025). "Heterogeneous Graph Neural Networks for Credulous Acceptance of Assumptions in ABA". in: *Proceedings of the Second International Workshop on Argumentation and Applications co-located with KR 2025.*
- Rapberger, Anna, **Fabrizio Russo**, Antonio Rago, and Francesca Toni (2025). "On Gradual Semantics for Assumption-Based Argumentation". In: *Proceedings of the 22nd International Conference on Principles of Knowledge Representation and Reasoning, KR 2025.*
- **Russo**, **Fabrizio** and Francesca Toni (2025). "Shapley-PC: Constraint-based Causal Structure Learning with a Shapley Inspired Framework". In: *Proceedings of the Fourth Conference on Causal Learning and Reasoning, CLeaR 2025.*
- **Russo**, **Fabrizio**, Anna Rapberger, and Francesca Toni (2024). "Argumentative Causal Discovery". In: *Proceedings of the 21st International Conference on Principles of Knowledge Representation and Reasoning, KR 2024.*
- Leofante, Francesco, Hamed Ayoobi, Adam Dejl, Gabriel Freedman, Deniz Gorur, Junqi Jiang, Guilherme Paulino-Passos, Antonio Rago, Anna Rapberger, **Fabrizio Russo**, Xiang Yin, Dekai Zhang, and Francesca Toni (Aug. 2024). "Contestable AI Needs Computational Argumentation". In: *Proceedings of the 21st International Conference on Principles of Knowledge Representation and Reasoning, KR 2024.*
- **Russo**, **Fabrizio** and Francesca Toni (2023). "Causal Discovery and Knowledge Injection for Contestable Neural Networks". In: *ECAI 2023 - 26th European Conference on Artificial Intelligence.* Volume 372. FAIA.
- **Russo**, **Fabrizio** (2023). "Argumentation for Interactive Causal Discovery". In: *Proceedings of the 32nd International Joint Conference on Artificial Intelligence, IJCAI 2023.*
- Rago, Antonio, **Fabrizio Russo**, Emanuele Albini, Pietro Baroni, and Francesca Toni (2021). "Forging Argumentative Explanations from Causal Models". In: *Proceedings of the 5th Workshop on Advances in Argumentation in Artificial Intelligence 2021 co-located with the 20th International Conference of the Italian Association for Artificial Intelligence (AIxIA 2021).*

Journal Articles

- Rago, Antonio, **Fabrizio Russo**, Emanuele Albini, Pietro Baroni, and Francesca Toni (2023). "Explaining Classifiers' Output with Causal Models and Argumentation". In: *Journal of Applied Logics — IfCoLog Journal of Logics and their Applications, Special Issue: Advances in Argumentation in AI* 10.3, pages 421–449.

SERVICE

Teaching

- **COMP70076 Ethics, Fairness and Explanation in AI**: Guest Lecture, *Causal Discovery for Trustworthy AI*, Imperial College London, London UK (Spring 2026)
- **Data Science In Action**: Guest Lecture, *Causal Discovery for Trustworthy AI*, Luiss Guido Carli, Rome Italy (Autumn 2025)
- **MSc in Fintech - Politecnico di Milano & UCL**: Guest Lecture, *From Credit Risk to Explainable AI Research*, UCL, London UK (Spring 2022)
- **COMP70050 Introduction to Machine Learning**: Tutorials and Marking, Imperial College London (2021, 2022)

Supervision

- Dhruv Himatsingka (MEng 2026): Bridging Gradual Semantics and Graph Neural Networks for Argumentative Causal Discovery
- Zihao Li (MSc 2025): Leveraging Large Language Models for Causal Discovery
- Yeva Hunanyan (MSc 2025): Exploring Semantics for Argumentative Causal Discovery
- Preesha Gehlot (MEng 2025): Graph Neural Networks for Approximate Argumentative Reasoning

Reviewing

- **Artificial Intelligence Journal**: Reviewer (2025, 2026)
- **KR**: Reviewer (2026)
- **IJCAI-ECAI**: Reviewer (2023, 2026)
- **AAAI**: Reviewer (2026)
- **NeurIPS**: Reviewer (2024, 2025)
- **ICLR**: Reviewer (2025)
- **ICML**: Reviewer (2022)
- **AISTATS**: Reviewer (2022, 2024); Top-Reviewer (2023)
- **AAMAS**: Reviewer (2023)

Organising

- **XAI Seminar Series @ Imperial**: Co-organiser (2021–present)
- **XAI-FIN at ICAIF**: Co-Chair (2025); Assistant Chair (2023, 2024); Workshop co-organiser (2022)
- **KCL–ICL Workshop on Causality**: Co-organiser (2025)

Invited Talks

- *CausAlly: Collaborative Causal Discovery for AI-Aided Decision-Making*. Minds and Machines Workshop, University of Queensland, Brisbane Australia (2025)
- *Causal Discovery for Trustworthy AI*. Lendable Journal Club, London UK (2025)
- *Causal Discovery for Trustworthy AI*. Lloyds Banking Group Summer School, London UK (2025)
- *Causal Discovery for Contestable and Explainable AI: From Contestable Neural Networks to Transparent Causal Discovery*. Robert Gordon University, Aberdeen UK (2024)
- *Argumentative Causal Discovery*. STAI CDT Student Seminar Series, London UK (2023)
- *Credit Risk Modelling: Data and Techniques Used in the UK Banking Industry*. Danmarks Nationalbank Conference, Copenhagen Denmark (2019)